

Enriching stock market data using Open AI API

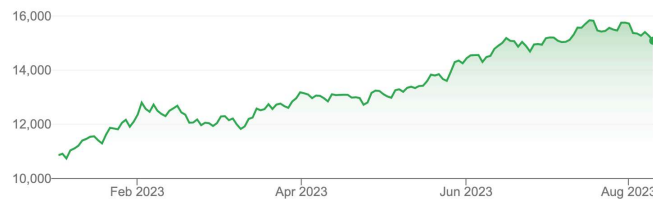
Market Summary > Nasdaq-100

15,101.71

+4,239.07 (39.02%) ↑ year to date

Aug 9, 5:15 PM EDT • Disclaimer

1D 5D 1M 6M YTD 1Y 5Y Max



Open	15,286.99	Low	15,077.91	52-wk high	15,932.05
High	15,288.76	Prev close	15,273.05	52-wk low	10,440.64

The Nasdaq-100 is a stock market index made up of 101 equity securities issued by 100 of the largest non-financial companies listed on the Nasdaq stock exchange. It helps investors compare stock prices with previous prices to determine market performance.

In this project you are provided with two CSV files containing Nasdaq-100 stock information:

- **nasdaq100_CA.csv**: contains information about companies in the index such as symbol, name, etc. For this analysis, only companies headquartered in California have been selected.
- **nasdaq100_price_change.csv**: contains price changes per stock across periods including (but not limited to) one day, five days, one month, six months, one year, etc.

As an AI developer, you will leverage the OpenAI API to classify companies into sectors and produce a summary of sector and company performance for this year, for the companies in the index that are headquartered in California.

CSV with Nasdaq-100 stock data

In this project, you have available two CSV files `nasdaq100_CA.csv` and `nasdaq100_price_change.csv`.

nasdaq100_CA.csv

```
symbol,name,headQuarter,dateFirstAdded,cik,founded
AAPL,Apple Inc., "Cupertino, CA",,0000320193,1976-04-01
ABNB,Airbnb,"San Francisco, CA",,0001559720,2008-08-01
ADBE,Adobe Inc., "San Jose, CA",,0000796343,1982-12-01
...
```

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nasdaq100_price_change.csv

```
symbol,1D,5D,1M,3M,6M,ytd,1Y,3Y,5Y,10Y,max
AAPL,-1.7254,-8.30086,-6.20411,3.042,15.64824,42.99992,8.47941,60.96299,245.42031,976.99441,139245.53954
ABNB,2.1617,-2.21919,9.88336,19.43286,19.64241,68.66902,23.64013,-1.04347,-1.04347,-1.04347,-1.04347
ADBE,0.5409,-1.77817,9.16191,52.0465,38.01522,57.22723,21.96206,17.83037,109.05718,1024.69214,251030.66399
ADI,0.9291,-4.03352,2.58486,3.65887,5.01602,17.02062,8.09735,63.42847,92.81874,286.77518,26012.63736
...
```

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```
# Start your code here!
import os
import pandas as pd
from openai import OpenAI

# Instantiate an API client
client = OpenAI(api_key=os.environ["OPENAI_API_KEY"])

# Continue coding here
# Use as many cells as you like
```

```
# Read both CSV files
nasdaq100_ca = pd.read_csv("nasdaq100_CA.csv")
price_change = pd.read_csv("nasdaq100_price_change.csv")

# Merge ytd column
nasdaq100_ca = nasdaq100_ca.merge(price_change[["symbol", "ytd"]], on="symbol", how="inner")

# Preview data
nasdaq100_ca.head()
```

...	↑↓	...	↑↓	name	...	↑↓	headQuarter	...	↑↓	dateFirstAd...	...	↑↓	...	↑↓	fou...	...	↑↓	y..	...
0		AAPL		Apple Inc.			Cupertino, CA						320193		1976-04-01			42.99	
1		ABNB		Airbnb			San Francisco, CA						1559720		2008-08-01			68.66	
2		ADBE		Adobe Inc.			San Jose, CA						796343		1982-12-01			57.22	
3		ADSK		Autodesk			San Rafael, CA						769397		1982-01-30			10.02	
4		AMAT		Applied Materials			Santa Clara, CA						6951		1967-11-10			55.46	

Rows: 5

Expand

```
# Loop each company
for company in nasdaq100_ca["symbol"]:

    # Build classification prompt
    prompt = f"""Classify company {company} into one of the following sectors. Answer only with the sector name:
Technology, Consumer Cyclical, Industrials, Utilities, Healthcare, Communication, Energy, Consumer Defensive, Real
Estate, Financial."""

    # Call OpenAI API
    response = client.chat.completions.create(
        model="gpt-4o-mini",
        messages=[{"role": "user", "content": prompt}],
        temperature=0.0
    )

    # Extract and save sector
    sector = response.choices[0].message.content
    nasdaq100_ca.loc[nasdaq100_ca["symbol"] == company, "sector"] = sector
```

```
# Count each sector
nasdaq100_ca["sector"].value_counts()
```

index	...	↑↓	...	↑↓
Technology			23	
Consumer Cyclical			8	
Healthcare			6	
Energy			1	

Rows: 4

Expand

```
# Build recommendation prompt
prompt = f"""Provide summary information about Nasdaq-100 stock performance year to date (YTD) of companies
headquartered in CA, recommending the two best sectors and two or more companies per sector.
Company data: {nasdaq100_ca}"""

# Call OpenAI API
response = client.chat.completions.create(
    model="gpt-4o-mini",
    messages=[{"role": "user", "content": prompt}],
    temperature=0.0
)
```

```
# Store and print results
stock_recommendations = response.choices[0].message.content
print(stock_recommendations)
```

Summary of Nasdaq-100 Stock Performance YTD for California Headquartered Companies

As of the current year, the performance of Nasdaq-100 companies headquartered in California shows a diverse range of returns across various sectors. The standout sectors based on year-to-date (YTD) performance are **Technology** and **Consumer Cyclical**.

1. Technology Sector

The Technology sector has shown remarkable performance, with several companies achieving significant YTD gains. Here are two of the best-performing companies in this sector:

- **Nvidia (NVDA)**: YTD Performance: **217.27%**
- **Meta Platforms (META)**: YTD Performance: **153.78%**
- **Advanced Micro Devices (AMD)**: YTD Performance: **82.46%**
- **Applied Materials (AMAT)**: YTD Performance: **55.46%**
- **Adobe Inc. (ADBE)**: YTD Performance: **57.23%**

The Technology sector is characterized by strong growth driven by advancements in AI, cloud computing, and semiconductor technologies.

2. Consumer Cyclical Sector

The Consumer Cyclical sector has also performed well, with several companies showing impressive YTD gains. Here are two notable companies:

- **Airbnb (ABNB)**: YTD Performance: **68.67%**
- **Netflix (NFLX)**: YTD Performance: **49.44%**
- **Electronic Arts (EA)**: YTD Performance: **1.02%**
- **Intuit (INTU)**: YTD Performance: **29.16%**

The Consumer Cyclical sector benefits from increased consumer spending and recovery in travel and entertainment,